

LAB RESOURCE UTILIZATION PLATFORM

Milestone 2— Project Documentation

Weeks 3 & 4

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Milestone	Milestone 2 (Week 3 & 4)
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Introduction

Milestone 1 of the Lab Resource Utilization Platform established the foundation of the system: user registration and authentication, role-based access control, institution and department management, equipment inventory, and basic equipment booking. With that foundation in place, the platform was able to record who the users are, what equipment exists, and which bookings have been made.

Milestone 2 builds directly on top of that foundation and shifts the focus from recording data to using data. The objective of this milestone is to convert raw booking and equipment records into meaningful operational intelligence — how heavily each instrument is actually used, when it sits idle, which equipment is in high demand, and how institutions can share expensive resources with one another instead of purchasing duplicates.

This document describes the scope, functionality, design, workflow, and technical implementation of Milestone 2 (Week 3 – 4).

Recap of Milestone 1

The following modules were completed in Milestone 1 and are consumed directly by Milestone 2. Milestone 2 does not duplicate any of this functionality; it extends it.

- User registration, login and JWT-based authentication.
- Role-Based Access Control across six roles.
- Institution and department master data.
- Equipment inventory management with status tracking.
- Equipment booking with approval and rejection workflow.

Milestone 2 therefore begins with an existing database containing real users, real equipment, and real bookings, and derives all of its analytics from that live data rather than from static or sample values.

Milestone 2 Objectives

- Calculate and display real-time equipment utilization from actual booking data.
- Provide a weekly utilization heatmap to visualise usage patterns across days.
- Automatically detect idle and under-utilised equipment.
- Enable inter-institution resource sharing so equipment can be requested across institutions.
- Allow external booking and access to shared equipment once a request is approved.
- Analyse utilization rates and identify high-demand equipment.
- Implement a waitlist mechanism so users are not blocked when equipment is already booked.
- Optimise booking by notifying waitlisted users when equipment becomes free.

Milestone 2 Scope

Milestone 2 consists of six functional tasks. Each task is implemented as an integrated part of the existing application rather than as an independent screen.

#	Milestone2Task	Pri mar y Role	SupportingRole
1	Real-ti me Equi pmen t Uti li zati on	Lab Manager / Department Head	Researcher / Student
2	Heatmap & Idle Time Detection	Lab Manager / Department Head	—
3	Int er-In sti tuti on Resou rce Shari ng	Lab Manager / Department Head	Insti tuti on Admin istrator
4	External Booking & Access	Researcher / Student	Lab Manager
5	Uti l iza tion Rate & Demand Analysis	Lab Manager / Department Head	Insti tuti on Admin istrator
6	Waitlist & Booking Optimization	Researcher / Student	Lab Manager

Module 1 – Real-Time Equipment Utilization

Purpose: to show how intensively each piece of laboratory equipment is being used, calculated from live booking records.

Functionality

- Display current operational status of every equipment item.
- Calculate total booked hours for each equipment item over a rolling seven-day period.
- Calculate the corresponding idle hours for the same period.
- Express usage as a utilization percentage.
- Restrict the data each user sees according to their institution and role.

Calculation Logic

The utilization period is a rolling window covering the current day and the six preceding days. For each equipment item the system sums the duration of all bookings that fall within this window and compares it against the total available hours in the same period.

Utilization%=(TotalBookedHours÷ TotalAvailableHours)× 100

Idle Hours =Total Available Hours – Total Booked Hours

All values are computed on the backend so that every dashboard, heatmap and report in the application uses one consistent definition of utilization.

Utilization Classification

UtilizationLevel	Ran ge	Interpretation
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UtilizationLevel	Ran ge	Interpretation
High	70%and above	Equipment is heavily used; a candidate for procurement of additional units.
Medium	30%– 69%	Equipment usage is healthy and balanced.
Low	Below 30%	Equipment is under-utilised; a candidate for sharing or reallocation.

Module 2 – Utilization Heatmap & Idle Time Detection

Purpose: to visualise when equipment is used during the week and to automatically surface equipment that is not being used at all.

He atm ap

- Usage is broken down day by day across the seven-day window (Monday to Sunday).
- Each day is assigned an intensity level derived from the hours booked on that day.
- The resulting grid allows managers to identify peak days and quiet days at a glance.
- Peak usage patterns can be used to plan maintenance during low-usage periods.

Idle Time Detection

- The system counts the number of days within the window on which an equipment item had no usage.
- Equipment with no recorded usage for the entire period is reported as fully idle.
- Idle equipment is highlighted so that it can be offered for sharing or moved to a department with higher demand.
- Idle detection uses the same booking data as utilization, ensuring both figures always agree.

Be ne fi t

Idle time detection directly supports the project objective of improving the utilization of expensive laboratory equipment. Instead of relying on manual observation, the platform identifies unused assets au to ma tic a lly.

Module 3 – Inter-Institution Resource Sharing

Purpose: to allow one institution to request the use of equipment owned by another institution, so that expensive instruments can be shared instead of duplicated.

Concept

A sharing request always involves two institutions. The institution that owns the equipment is the providing institution, and the institution that needs the equipment is the requesting institution. The platform records the request, routes it for approval, and tracks its outcome.

Functionality

- View the catalogue of equipment that has been made shareable by other institutions.
- Raise a request for equipment belonging to another institution.
- View all incoming and outgoing sharing requests.
- Approve or reject an incoming sharing request.
- Track the status of every request through its lifecycle.

Validation Rules

The following rules are enforced by the backend so that invalid sharing data cannot be created:

- The requesting institution and the providing institution must be different.
- The selected equipment must genuinely belong to the providing institution.
- A user may only raise a request on behalf of their own institution.
- Only authorised roles may approve or reject a request.

Request Status Lifecycle

Status	Meaning
P E N D I N G	The request has been submitted and is awaiting a decision from the providing institution.
APPROVED	The providing institution has granted access; the equipment may now be booked externally.
R E J E C T E D	The providing inst itution has declined the request.

Module 4 – External Booking & Access

Purpose: to make an approved sharing arrangement usable in practice, by allowing users from the requesting institution to book equipment belonging to the providing institution.

Functionality

- Once a sharing request is approved, the shared equipment becomes visible to the requesting institution.
- Users of the requesting institution can create bookings against that equipment.

- External bookings follow the same approval workflow as internal bookings.
- The owning institution retains full approval authority over its own equipment.
- Usage generated through external bookings is included in the owning institution's utilization figures.

Booking Workflow

1. The user selects the equipment and the required date and time.
2. The system validates the requested slot against existing bookings.
3. The booking is created with the status Pending Approval.
4. The Lab Manager or Department Head approves or rejects the booking.
5. On approval the booking status becomes Confirmed and the equipment status is updated.
6. After the booking period ends the booking is marked Completed and usage is recorded.

Module 5 – Utilization Rate & Demand Analysis

Purpose: to summarise utilization across the institution and to identify which equipment is in greatest demand.

Summary Metrics

Metric	Description
Total Equipment	Number of equipment items visible to the logged-in user.
Average Utilization	Mean utilization percentage across all visible equipment.
Most Requested Equipment	Equipment attracting the highest number of booking requests.
Highest Utilization Equipment	Equipment with the greatest proportion of booked hours.
Lowest Utilization Equipment	Equipment with the least usage; a candidate for sharing.

Demand Analysis

- Ranks equipment by booking volume to reveal genuine demand.
- Highlights equipment where demand consistently exceeds availability.
- Supports procurement planning by distinguishing high-demand from under-used assets.
- Provides Department Heads and Institution Administrators with evidence for resource decisions.

Role-Based Data Scope

Analytics are scoped so that each role sees only the data it is entitled to. Manager-tier roles see data belonging to their own institution, while the System Administrator can view data across the entire platform. This scoping is enforced on the backend, not merely hidden in the user interface.

Module 6 – Waitlist & Booking Optimization

Purpose: to ensure that demand is captured rather than lost when equipment is already fully booked.

Functionality

- A user who cannot book equipment may join a waitlist for the required time slot.
- The requested start and end time are stored with the waitlist entry.
- Users can view the status of their own waitlist entries.
- Users can cancel a waitlist entry that is no longer required.
- When equipment becomes available, the waitlisted user is notified and can confirm or decline.
- Priority handling is supported so that urgent requirements can be escalated.

Waitlist Status Lifecycle

Status	Meaning
WAITING	The user is queued and waiting for the equipment to become available.

Status	Meaning
NOTIFIED	The equipment has become available and the user has been informed.
F ULFI LLED	The user accepted the offered slot and a booking was created.
CA NCE LLE D	The waitlist entry was cancelled by the user or is no longer valid.

Booking Optimization Benefit

The waitlist mechanism converts cancelled and completed bookings into new opportunities. Equipment that would otherwise sit idle after a cancellation is offered immediately to the next user in the queue, which increases overall utilization and reduces scheduling conflicts.

Role-Based Dashboards

Milestone 2 functionality is presented through role-specific dashboards. Each role sees only the components relevant to its responsibilities.

Researcher / Student

- MyBookingsandUpcomingReservations
- EquipmentAvailabilityandBookingHistory
- UsageSummary
- WaitlistStatus
- Notifications

Lab Manager

- Equipment Utilization and Utilization Heatmap
- Booking Statistics and No-Show Rates
- High-Demand Equipment and Idle Equipment
- Resource Sharing and Sharing Requests
- Booking Approvals

Department Head

- Department Equipment and Utilization
- Utilization Heatmap and Demand Analysis
- Booking Statistics
- Resource Sharing Requests
- DepartmentReports

Institution Administrator

- Organization-wide equipment utilization
- Cross-department and inter-institution resource sharing overview
- Department and user management
- Institution-wide reports

Lab Technician

- Equipment status and availability
- Assigned tasks
- Equipment downtime visibility

System Administrator

- Platform-wide access to all institutions and departments
- User and role management

- Access management and system monitoring

Recommended Frontend Navigation

Role	Nav i gati o n It ems
Researcher / Student	Dashboard Equipment My Bookings Booking History Waitlist Notifications Profile
Lab Technician	Dashboard Equipment Assigned Tasks Maintenance Maintenance History
Lab Manager	Dashboard Equipment Bookings Utilization Heatmap Idle Equipment Demand Analysis Resource Sharing Requests Reports
Department Head	Dashboard Department Equipment Utilization Heatmap Demand Analysis Bookings Resource Sharing Reports
Insti tuti on Admin istrator	Dashboard Departments Equipment Utilization Resource Sharing Users Analytics Reports
System Admi ni stra tor	Dashboard Users / Roles Access Management System Monitoring Co nf ig ura t io n

Statuses Used in Milestone 2

Equipment Status

- Available
- Booked
- In Use
- Under Maintenance
- Out of Service
- Retired

Booking Status

- Pending Approval
- Confirmed
- Completed
- Cancelled
- Rejected

Waitlist Status

- WAITING
- NOTIFIED
- FULFILLED
- CANCELLED

Sharing Request Status

- PENDING
- APPROVED
- R E J E C T E D

API Endpoints Implemented in Milestone 2

All endpoints are secured using JWT authentication and are further protected by role-based authorization on the backend.

Utilization & Demand Analysis

Met ho d	Endpoint	Description
GET	/api/utilization	Returns per-equipment utilization, idle hours and h ea tmap va lues.
GET	/ ap i/ ut i li zat i o n/s um mary	Returns aggregated utilization and demand metrics.

Resource Sharing

Met ho d	Endpoint	Description
GET	/ ap i/ reso urc e- s hari ng /req uest s	Lists sharing requests relevant to the user's institution.
POST	/ ap i/ reso urc e- s hari ng /req uest s	Creates a new inter-institut ion sharing request.
PUT	/ ap i/ re so urc e - s hari ng /re q ue st s/ {id}	Approves or rejects a sharing request.

Booking

Met ho d	Endpoint	Description
GET	/api/bookings	Lists bookings visible to the user.
GET	/api/bookings/{id}	Retrieves a single booking.
PUT	/api/bookings/{id}	Updates an existing booking.
DE LE TE	/api/bookings/{id}	Cancels a booking.
PUT	/ ap i/ b oo ki ng s/ {i d}/ ap pro ve	Ap p rov es a pen di ng b ooki ng .
PUT	/ ap i/ b oo ki ng s/ {i d}/ rejec t	R eject s a p en di ng b ooki ng .
PUT	/api/bookings/{id}/complete	Marks a booking as completed and records usage.

Waitlist

Met ho d	Endpoint	Description
POST	/api/waitlist	Joins the waitlist for an equipment item and ti me slot.
GET	/api/waitlist/my	Lists the logged-in user's waitlist entries.
GET	/api/waitlist/equipment/{equipmentId}	Lists the waitlist queue for an equipment item.
POST	/api/waitlist/{id}/decide	Accepts or declines an offered slot.
DE LE TE	/api/waitlist/{id}	Cancels a waitlist entry.

Database Extensions for Milestone 2

Milestone 2 reuses the Milestone 1 schema and adds only the tables required by the new functionality. No existing table was redesigned.

New Entities

Entity	Purpose	KeyFields
Waitlist	Stores queued demand for equipment that is already booked.	waitlist_id, user_id, equipment_id, requested_start_time, requested_end_time, waitlist_status, is_priority, queue_date
Resource Sharing Request	Records inter-institution equipment sharing requests.	id, equipment_id, sender_institution_id, receiver_institution_id, request_date, status
Notification	Delivers booking, waitlist and sharing alerts to users.	notification_id, user_id, message, type, is_read, created_at

Entity Relationships

- One User can create many Waitlist entries.
- One Equipment can have many Waitlist entries.
- One Equipment can be the subject of many Resource Sharing Requests.
- One Institution can send many Resource Sharing Requests.
- One Institution can receive many Resource Sharing Requests.
- One User can receive many Notifications.
- Utilization is derived from existing Booking and Equipment records and is not stored as a separate table.

Design Note

Utilization, idle hours, heatmap values and demand metrics are calculated at request time from existing booking records rather than being stored in the database. This guarantees that the figures shown on every dashboard always reflect the current state of the data and cannot become stale.

Database Schema & ER Diagram

The diagram below shows the Milestone 2 database schema. The tables added in this milestone are highlighted, and the derived utilization layer is shown to indicate that it is calculated rather than stored.

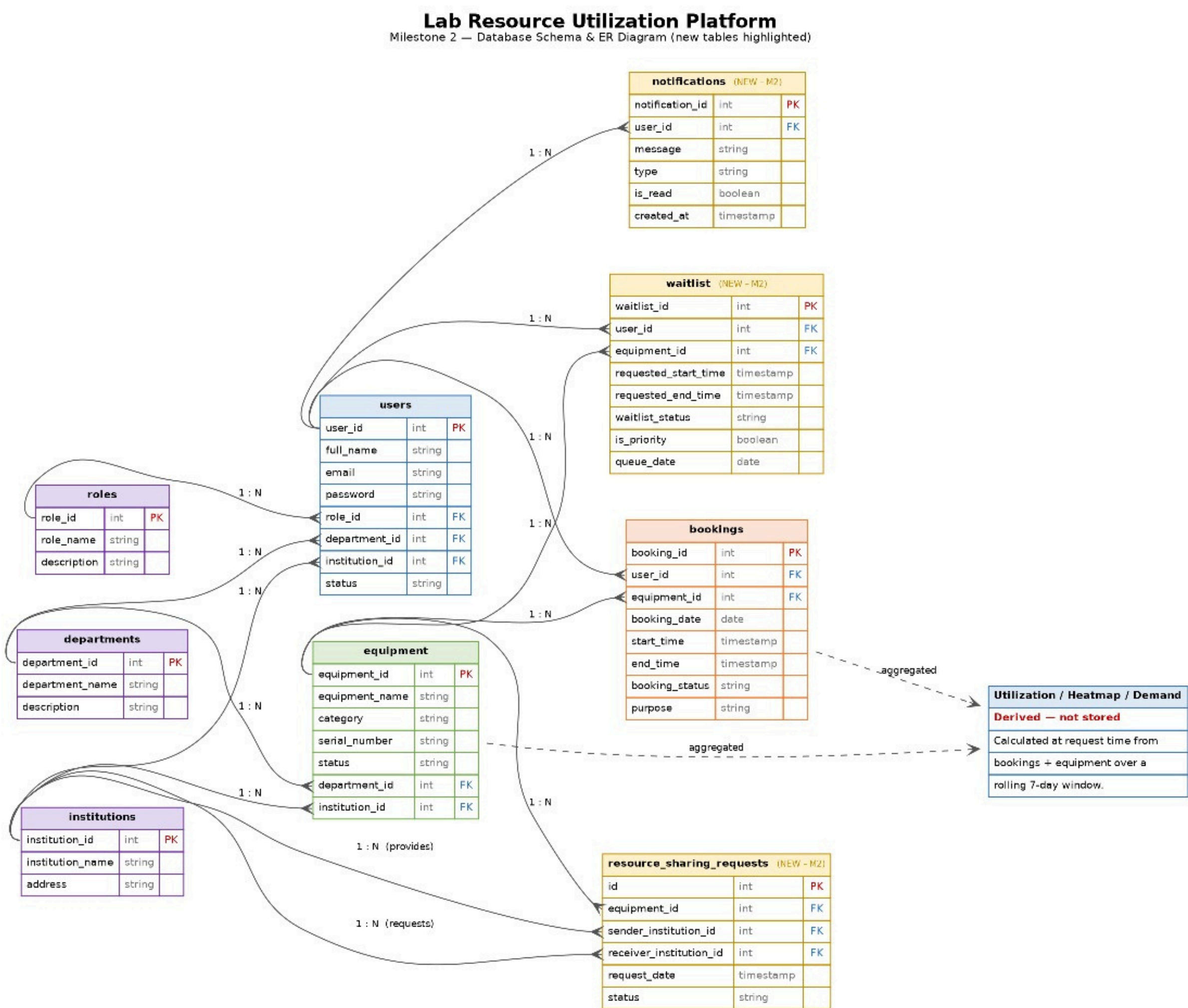


Figure 2— Milestone 2 Database Schema & ER Diagram

Milestone 2 Complete User Flow

Milestone 2 is implemented as one integrated workflow rather than six independent screens.

7. The researcher searches for laboratory equipment.
8. The researcher checks the availability of the selected equipment.
9. If the equipment is available, the researcher submits a booking request.
10. The Lab Manager or Department Head approves or rejects the request.
11. On approval, the equipment status is updated according to the booking state.
12. When the booking period ends, the booking is completed and usage data is recorded.
13. The system recalculates the utilization percentage from the recorded usage.
14. The Lab Manager dashboard, heatmap and demand analysis are updated automatically.
15. If the equipment is unavailable, the researcher joins the waitlist instead.

16. When the equipment becomes free, the waitlisted user is notified and may confirm the slot.
17. If suitable equipment is not available within the institution, a sharing request is raised to another institution.
18. Once the sharing request is approved, the equipment can be booked externally and its usage contributes to the owning institution's utilization.

Milestone 2 Workflow Diagram

The diagram below shows the complete Milestone 2 workflow, including the booking path, the waitlist path, the inter-institution sharing path, and the utilization calculation that follows from recorded usage.

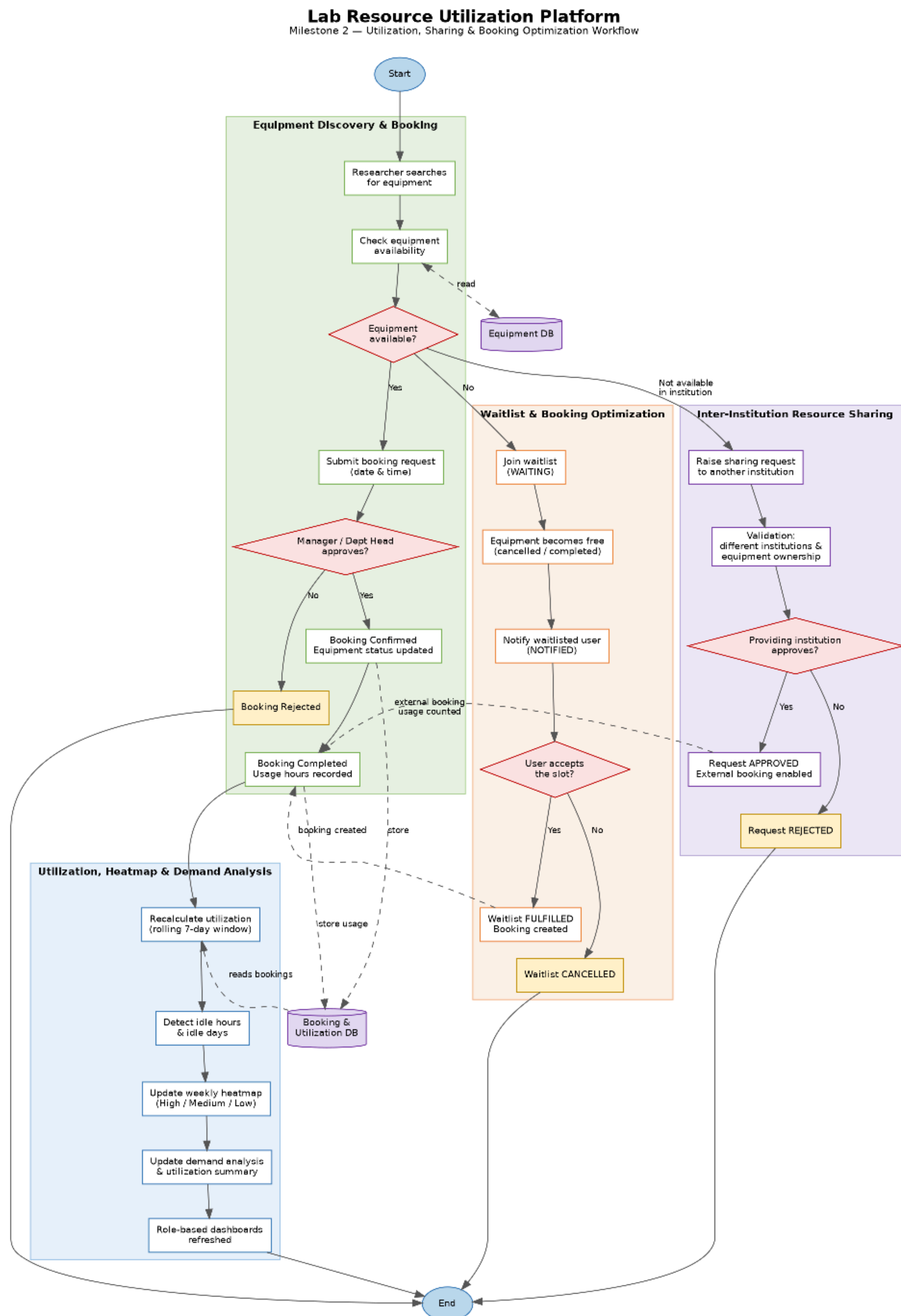


Figure 1 — Milestone 2 Workflow / Process Flow

Functional Requirements

Utilization Monitoring

- The system shall calculate equipment utilization from actual booking records.
- The system shall display utilization percentage, booked hours and idle hours.
- The system shall present a weekly utilization heatmap.
- The system shall identify idle equipment automatically.

Resource Sharing

- The system shall allow an institution to request equipment from another institution.
- The system shall prevent a sharing request between the same institution.
- The system shall validate that the equipment belongs to the providing institution.
- The system shall allow authorised roles to approve or reject sharing requests.

Booking and Waitlist

- The system shall allow users to book available equipment.
- The system shall allow authorised roles to approve, reject or complete bookings.
- The system shall allow users to join a waitlist when equipment is unavailable.
- The system shall notify waitlisted users when equipment becomes available.
- The system shall allow users to cancel their own waitlist entries.

Demand Analysis

- The system shall identify the most requested equipment.
- The system shall identify the highest and lowest utilised equipment.
- The system shall present an average utilization figure for the institution.

Non-Functional Requirements

Security

- All Milestone 2 endpoints require a valid JWT token.
- Authorization is enforced on the backend using role-based rules.
- Users can only access utilization and sharing data belonging to their own institution, except the System Administrator.

Performance

- Utilization calculations are performed on the backend to avoid duplicated logic in the frontend.
- Aggregations are computed in a single pass over the relevant booking records.

Reliability

- Empty datasets are handled gracefully and do not cause errors.

- Division-by-zero conditions are guarded when no available hours exist.
- Invalid or missing optional values are handled without failure.

Usabil ity

- Utilization levels are colour-coded as high, medium and low for quick interpretation.
- Dashboards display only the components relevant to the logged-in role.
- Status values are presented consistently across all screens.

Testing and Validation

Milestone 2 functionality was validated using both the application interface and direct API testing.

Test Area	ValidationPerformed
Uti l iza tion a ccu racy	Created a booking and confirmed that booked hours, idle hours and utilization p ercentag e chan ged accordi ng l y.
Heatmap	Verified that usage appears on the correct day of t he week.
Idle detection	Confirmed that equipment with no bookings is reported as idle for the full p erio d .
Resource sharing validation	Confirmed that same-institution requests and mismatched equipment are rejected by the backend.
Sh arin g approval	Veri fi ed that onl y authori sed rol es can approve or rej ect a request .
Bookingworkflow	Verified the pending, confirmed, completed, cancelled and rejected transitions.
Waitlist	Veri fi ed j oi ni ng, vi ewin g, deci ding a nd cancell i ng a wa itl ist entr y.
Role-basedaccess	Confirmed that unauthorised roles receive an access-denied response rather th an data .

Expected Outcome

On completion of Milestone 2, the Lab Resource Utilization Platform is able to measure and improve how laboratory equipment is used, rather than simply recording bookings.

- Real-time visibility of equipment utilization across the institution.
- Automatic identification of idle and under-utilised equipment.
- Evidence-based demand analysis to support procurement decisions.
- A working inter-institution sharing workflow that reduces duplicate purchases.
- External booking of shared equipment with the owning institution retaining control.
- A waitlist mechanism that captures unmet demand and increases overall utilization.
- Consistent, backend-enforced role-based access across all new functionality.

Conclusion

Milestone 2 extends the Lab Resource Utilization Platform from a record-keeping system into a

resource-optimisation system. By deriving utilization, idle time and demand directly from live booking data, and by enabling institutions to share equipment and queue unmet demand, the platform now delivers the analytical foundation required for the maintenance, cost management, analytics and reporting modules planned for Milestone 3.

UI Screens

The following screens show the Milestone 2 features running on the live deployed application (frontend hosted on Render, connected to the Spring Boot backend and the PostgreSQL/Neon database).

The login screen authenticates the user against the users table and issues a JWT token on success. The role and institution encoded in this token determine which dashboard the user is redirected to and which data they are permitted to see, as described in the Role-Based Dashboards section.

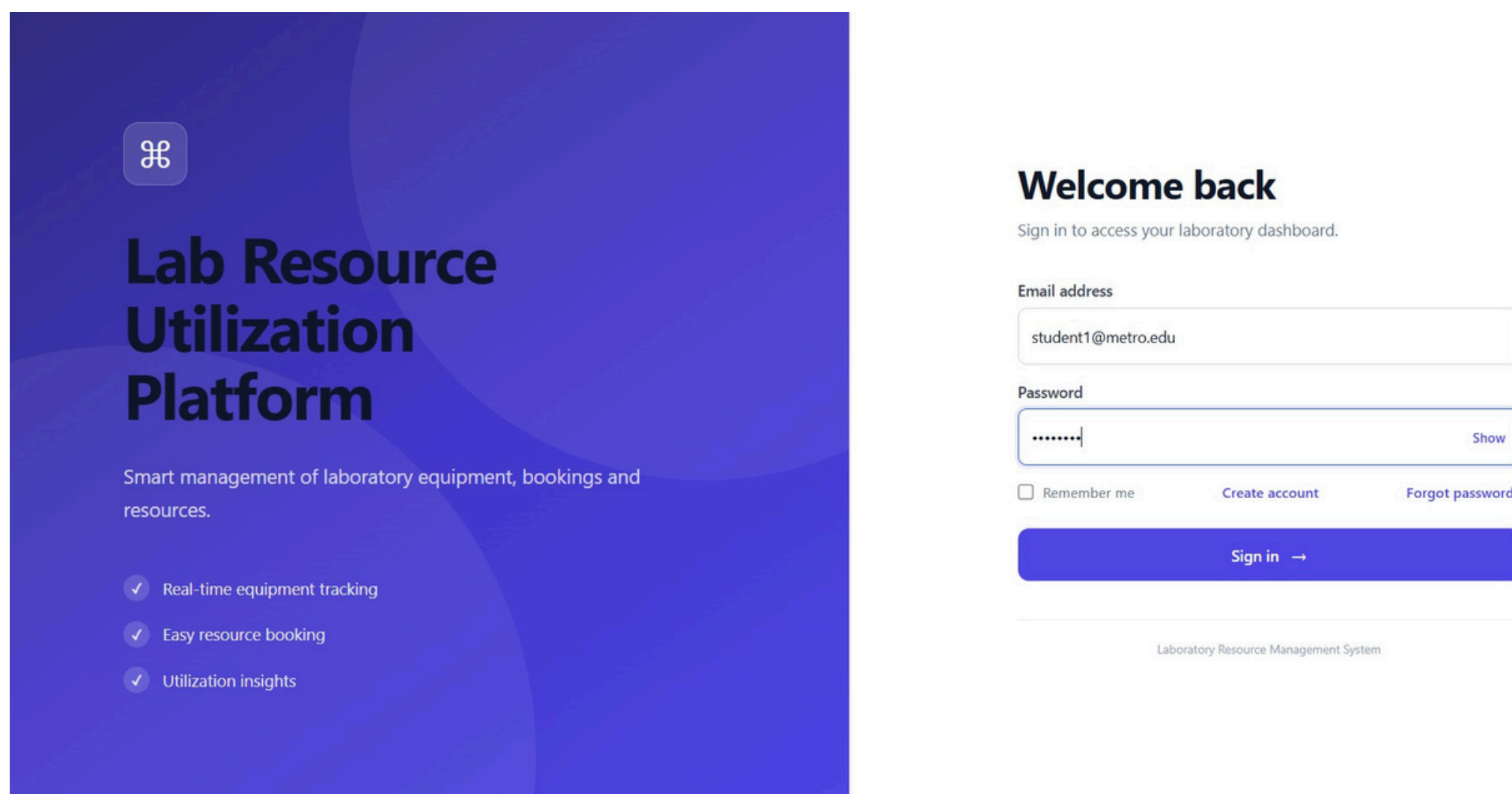


Figure3 — Login Page

The Lab Manager dashboard summarises the equipment under the manager's institution: total equipment, currently available equipment, and currently booked equipment. The Equipment Utilization table below it lists used hours, idle hours and utilization percentage per equipment item for the rolling seven-day window described in Module 1, together with a High / Medium / Low category and the number of idle days.

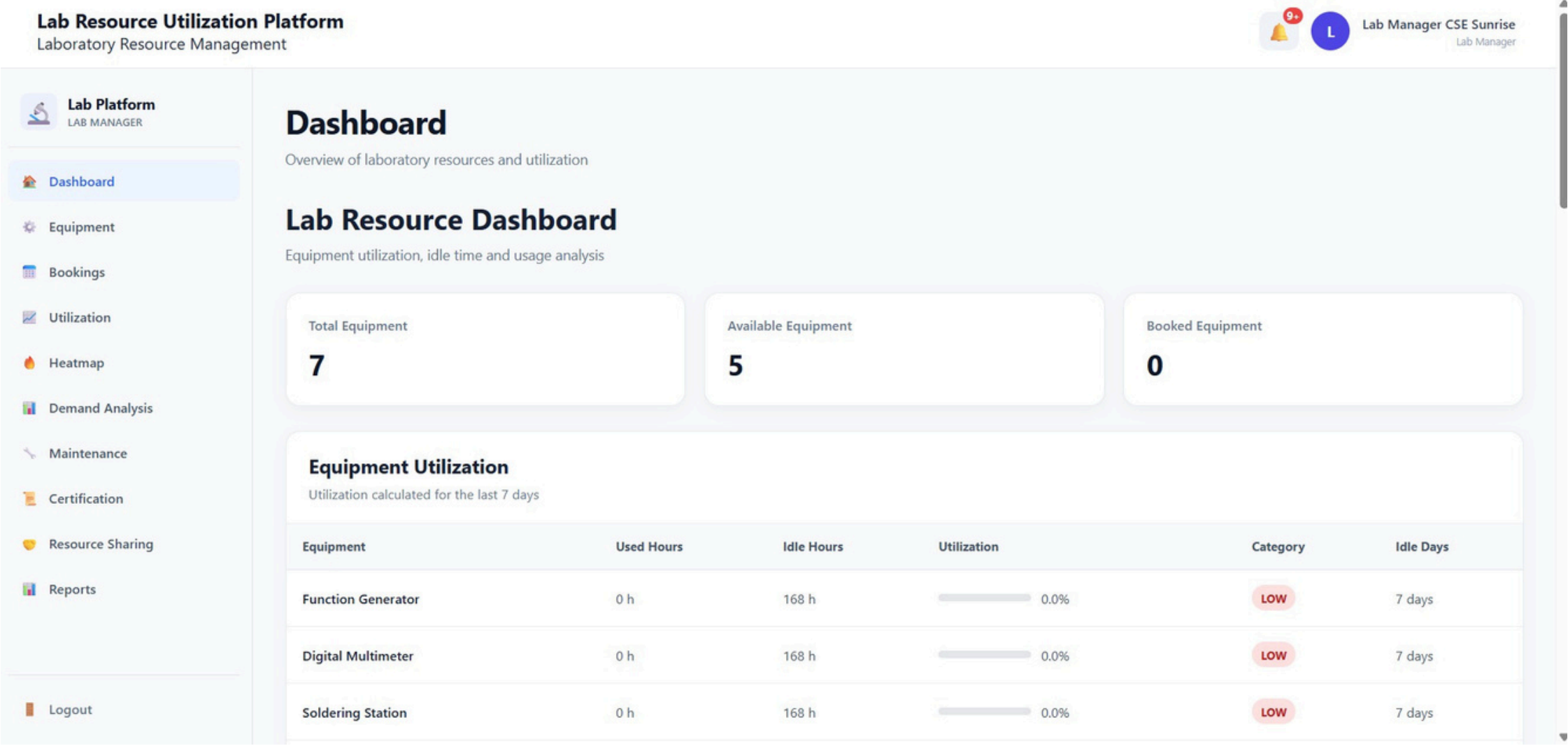


Figure 4 — Lab Manager Dashboard (Equipment Utilization Overview)

The Equipment Utilization page presents the same seven-day utilization calculation as a dedicated, filterable view with summary cards for total equipment, average utilization, total used hours and total idle hours, along with a per-equipment utilization chart. This is the primary screen through which a Lab Manager monitors real-time equipment usage, as specified in Module 1.

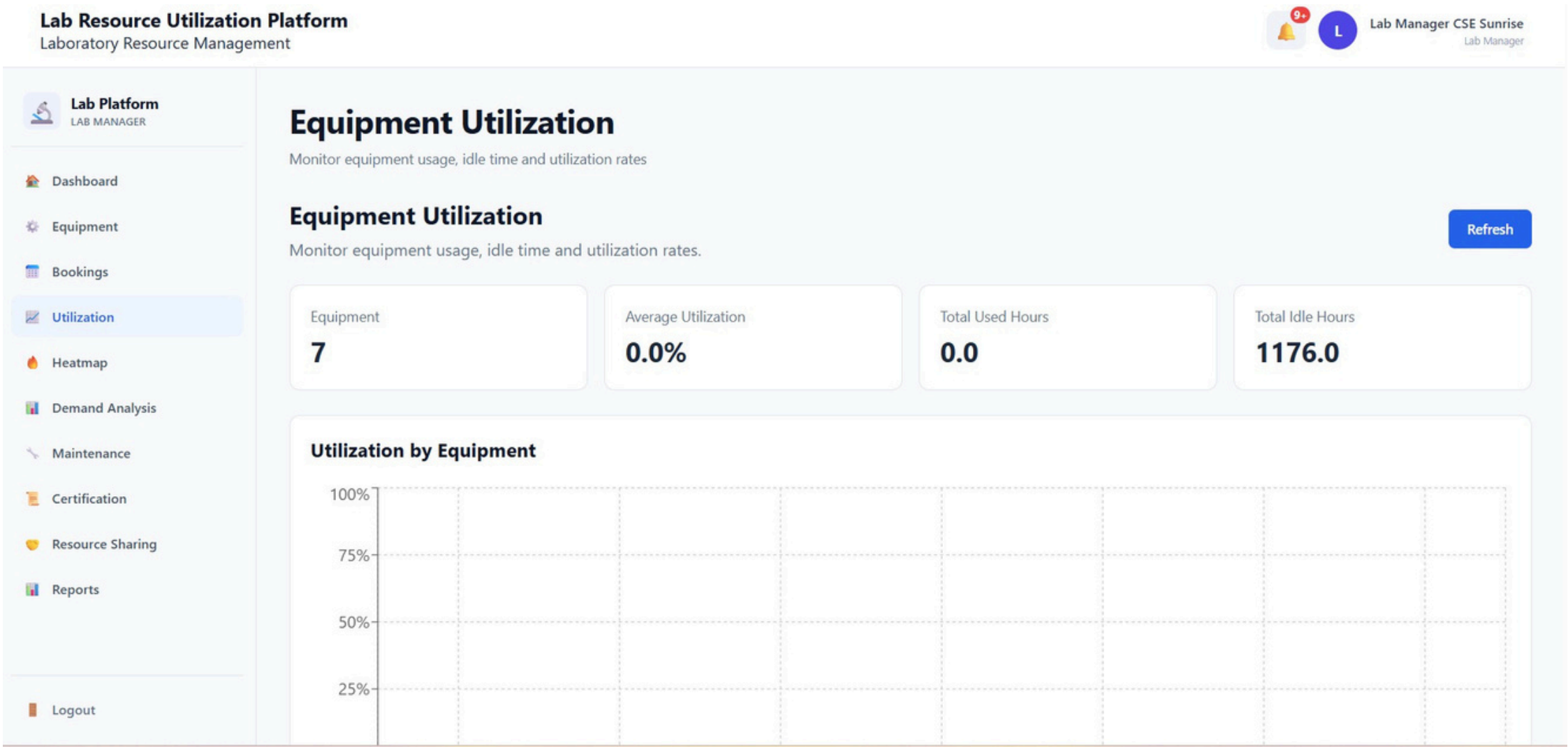


Figure 5— Equipment Utilization

The Utilization Heatmap breaks usage down by day of the week for every equipment item and colour-codes each cell as High, Medium, Low or Idle, exactly as defined in Module 2. The Idle Days column on the right counts how many of the seven days had no recorded usage at all, which is how the platform automatically flags equipment as a candidate for resource sharing or reallocation.

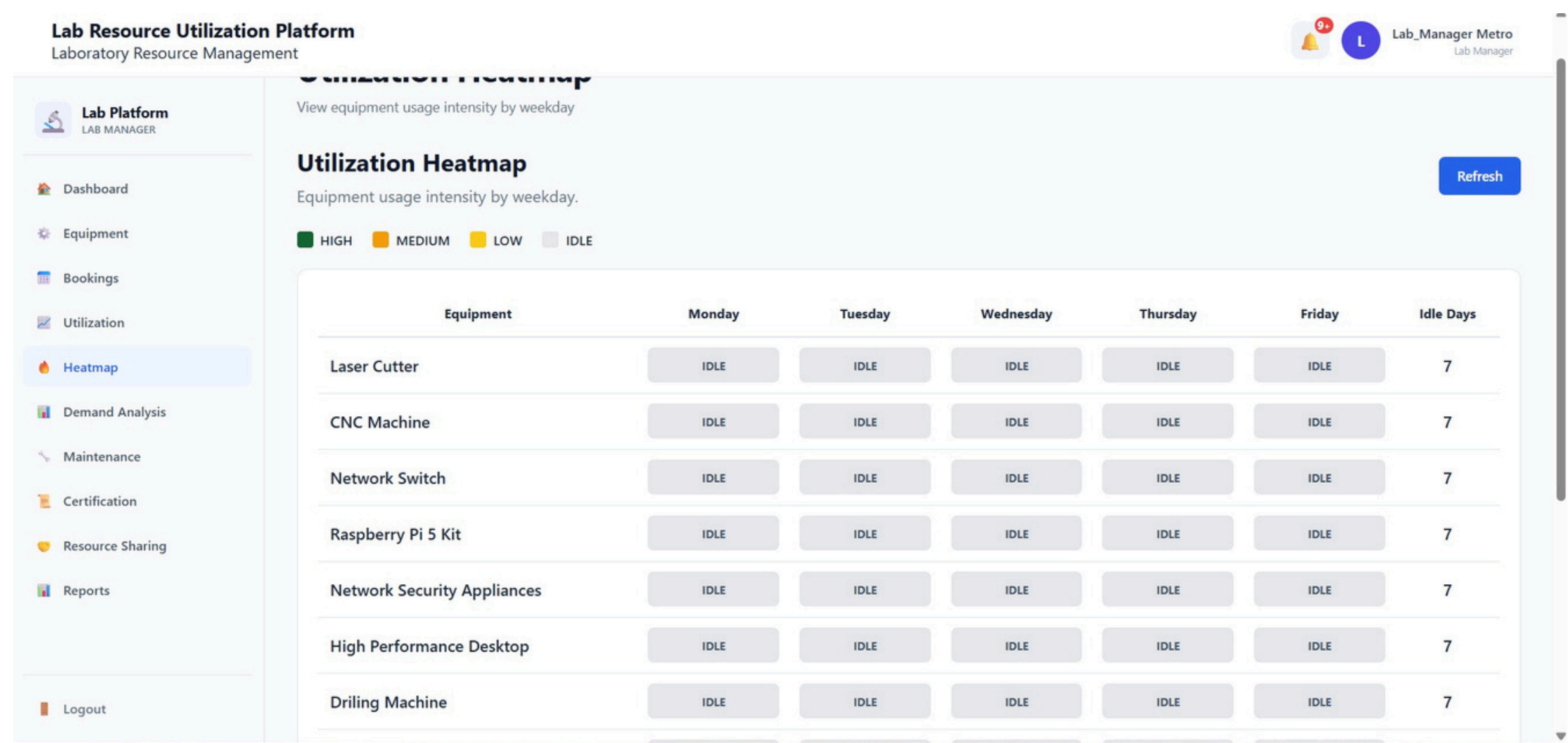


Figure6 – UtilizationHeatmap

The Demand Analysis page aggregates booking activity across the institution to identify the most requested equipment and the equipment with the highest and lowest utilization, matching the summary metrics defined in Module 5. This gives Lab Managers and Institution Administrators evidence to support procurement and reallocation decisions.

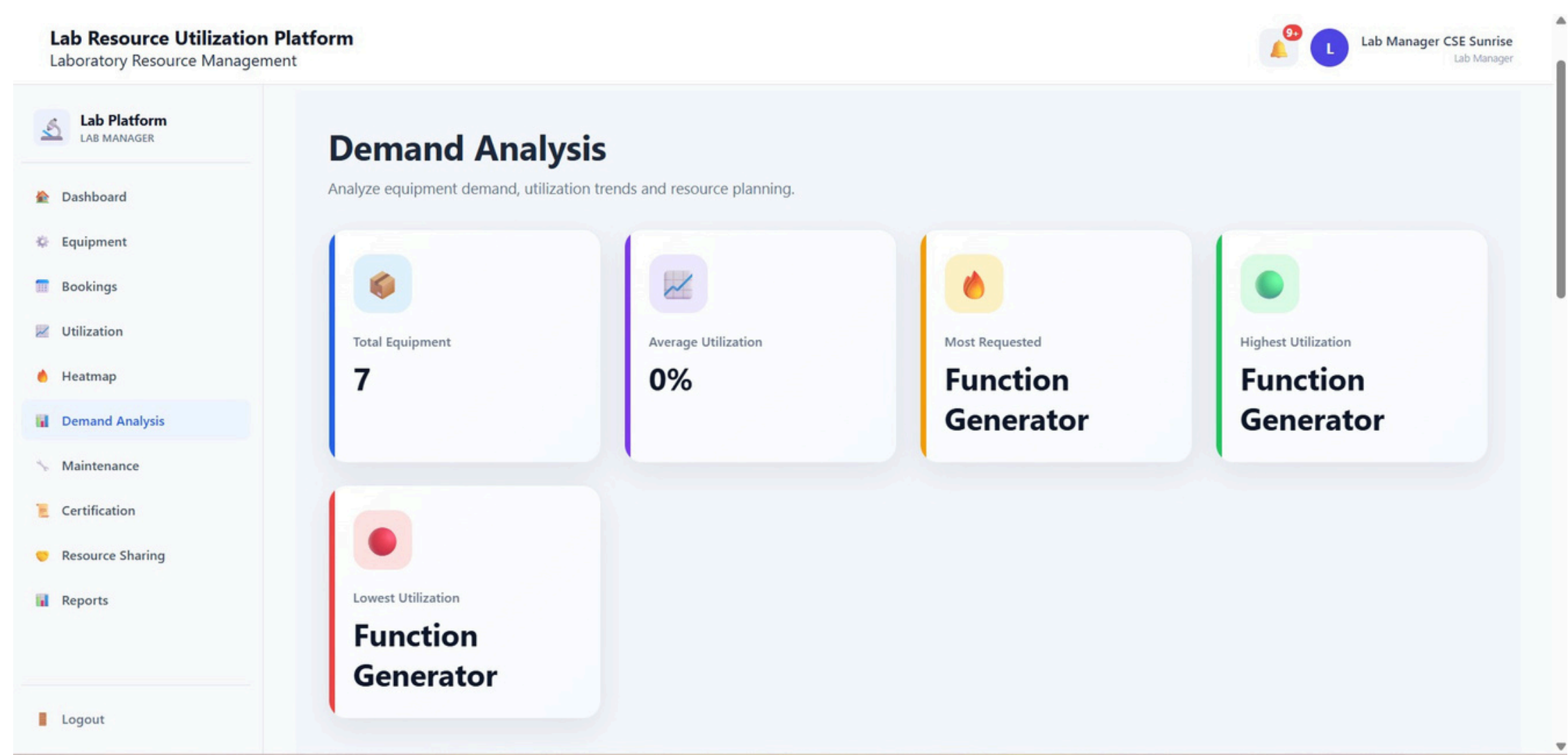


Figure 7 – Demand Analysis

The Inter-Institution Resource Sharing page lets a Lab Manager raise a request for equipment owned by another institution and shows the status of every incoming and outgoing request. The Approve / Reject actions and the PENDING, APPROVED and REJECTED statuses visible here correspond directly to the sharing request lifecycle defined in Module 3.

Lab Resource Utilization Platform
Laboratory Resource Management

Lab Platform
LAB MANAGER

Dashboard

Equipment

Bookings

Utilization

Heatmap

Demand Analysis

Maintenance

Certification

Resource Sharing

Reports

Logout

Inter-Institution Resource Sharing

Send Sharing Request

Institution you're requesting equipment from:

National Institute of Technology

Equipment:

-- Select Equipment --

Send Request

Incoming Requests (11)

Outgoing Requests (2)

Equipment	Sender Institution	Receiver Institution	Status	Actions
Function Generator	Sunrise Institute of Technology	Metro College of Engineering	PENDING	ApproveReject
Function Generator	Sunrise Institute of Technology	Metro College of Engineering	PENDING	ApproveReject
Function Generator	Sunrise Institute of Technology	Metro College of Engineering	PENDING	ApproveReject
Function Generator	Sunrise Institute of Technology	Metro College of Engineering	PENDING	ApproveReject
Function Generator	Sunrise Institute of Technology	Metro College of Engineering	PENDING	ApproveReject
Arduino Development Kit	Sunrise Institute of Technology	Metro College of Engineering	PENDING	ApproveReject
Arduino Development Kit	Sunrise Institute of Technology	Metro College of Engineering	APPROVED	
Arduino Development Kit	Sunrise Institute of Technology	Metro College of Engineering	REJECTED	
Arduino Development Kit	Sunrise Institute of Technology	Metro College of Engineering	APPROVED	
Arduino Development Kit	Sunrise Institute of Technology	Metro College of Engineering	REJECTED	
Arduino Development Kit	Sunrise Institute of Technology	Metro College of Engineering	REJECTED	

Figure 8 – Inter-Institution Resource Sharing

The My Waitlist page, shown here from a Student account, lists the equipment the user is queued for along with the requested start and end time and the current status. This corresponds to the WAITING state in the waitlist lifecycle defined in Module 6, and is the screen a student uses to track or cancel their place in the queue.

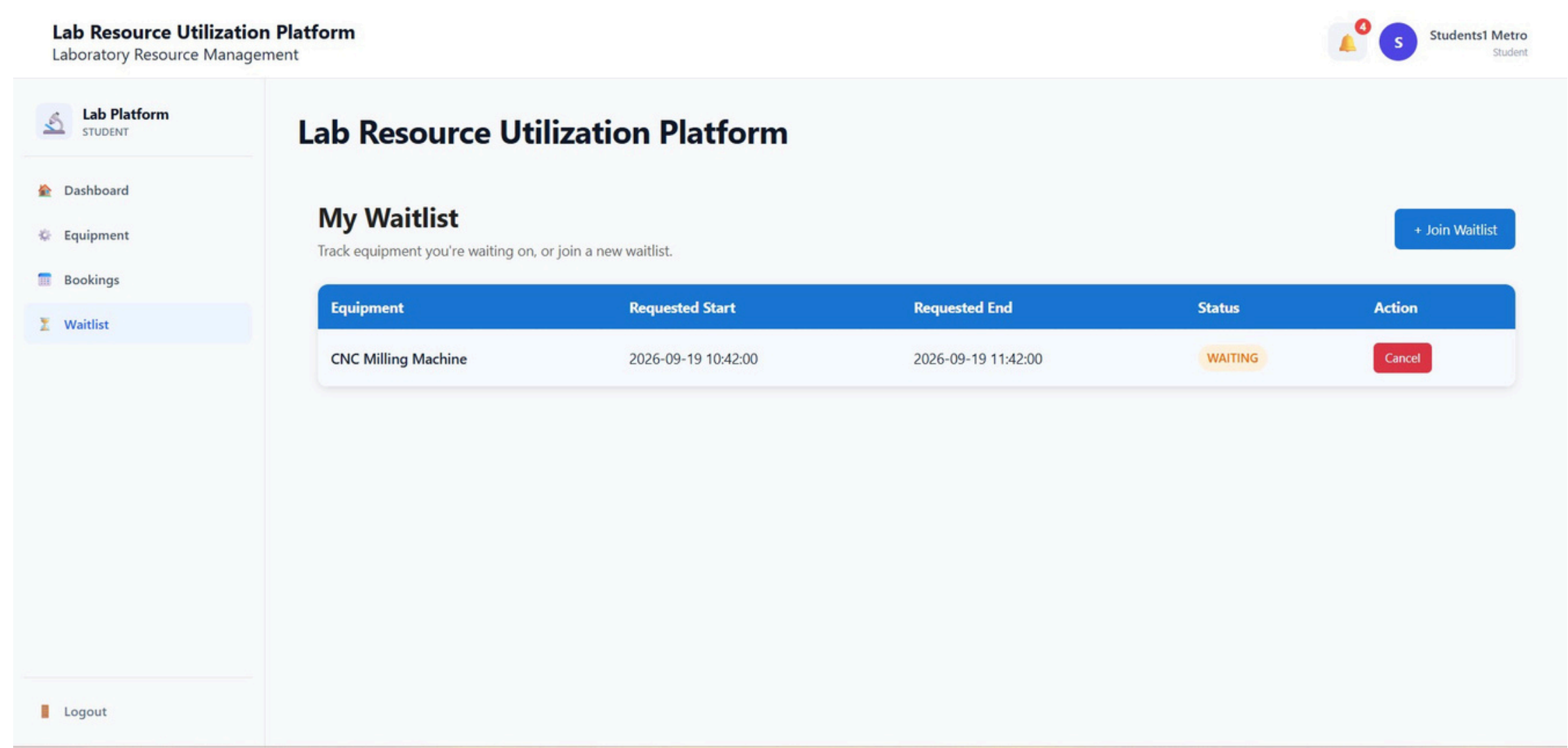


Figure9 – Student Waitlist View